

Full Technical Documentation

AI-Powered Smart Education Platform & Progressive Web App (PWA)

Comprehensive architectural reference, Progressive Web App notification download specification, algorithm definitions, backend REST APIs, and deployment configurations.

PWA PRODUCTION APP

<https://skillnex.vercel.app>

GITHUB REPOSITORY

[ksivakumarkalaiselvan/skillnex](https://github.com/ksivakumarkalaiselvan/skillnex)

ARCHITECTURE VERSION

v2.4.0 (PWA + Notification Engine)



1. Executive System Overview

SKILLNEX is an AI-powered Smart Education ecosystem designed to deliver dynamic, personalized, and measurable learning experiences. By combining a client-driven progressive web application architecture, local LLM inference engines, subject-isolated assessment metrics, and Google Sheets cloud databases, SKILLNEX eliminates learning friction and accelerates skill acquisition.

Core Architectural Pillars

Unified Platform

- **Notification-Driven PWA Installation:** Seamless 1-click app download triggered directly via native browser notifications.
- **Data-Driven Student Dashboard:** Focused on 4 deterministic metrics (Student Name, Overall Skill Score, Active Courses, Best Subject).
- **Single-Subject AI Study Planner:** Generates targeted 5-day remediation workflows specifically for identified weak subjects.
- **Isolated Subject Assessments:** Standardized multi-subject quiz engine with exact score and skill level classification algorithms.
- **Local Ollama AI Integration:** Offline, zero-latency LLM workspace supporting local model inference (`qwen2.5` , `llama3`).
- **Teacher Copilot & Risk Warning Matrix:** Automated early warning detection for struggling students across classrooms.

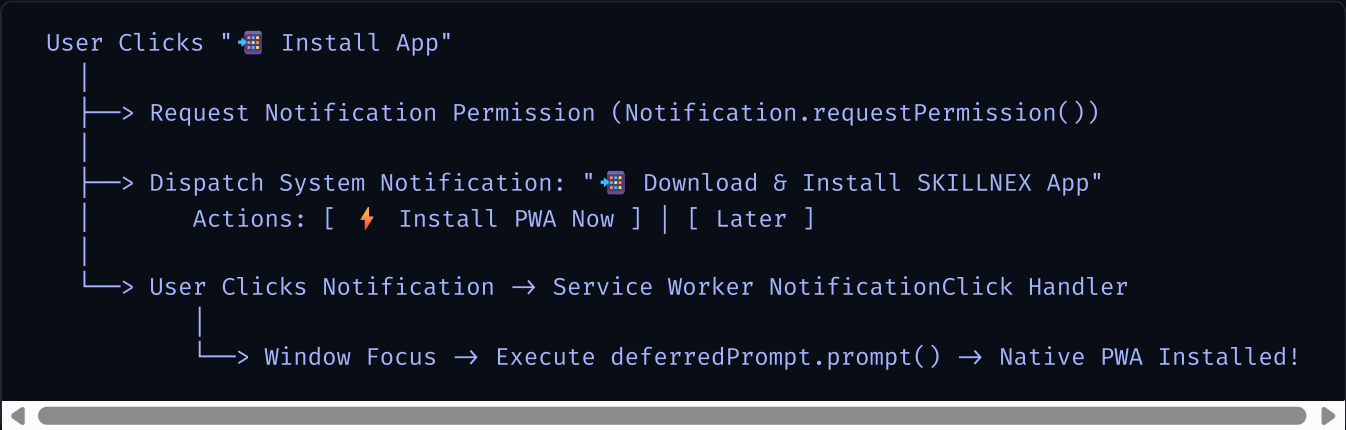


2. Downloadable PWA & Notification System Architecture

SKILLNEX incorporates a modern Progressive Web App (PWA) layer designed to convert the web application into an offline-capable, desktop/mobile downloadable app through an interactive notification flow.

PWA Components & Specification			Service Worker v1.0
Component File	Path	Technical Function	
Web App Manifest	frontend/manifest.json	Defines app identity, branding colors (#0f172a , #6366f1), standalone display mode, multi-resolution icons (192x192, 512x512, maskable), and dashboard shortcuts.	
Service Worker	frontend/sw.js	Pre-caches static assets, executes Stale-While-Revalidate caching for static files, Network-First strategy for /api/ endpoints, and handles push & notificationclick events.	
PWA Installer Manager	frontend/js/pwa-installer.js	Captures beforeinstallprompt , dispatches native OS install notifications, controls in-app download badges, and executes deferredPrompt.prompt() .	
PWA Glassmorphism Styles	frontend/css/pwa.css	Provides glassmorphism modal styles, notification toasts, floating status badges with live CSS pulsing keyframe animations.	
App Branding Icons	frontend/icons/	Generated vector & raster icon assets(icon-192.png , icon-512.png , icon-maskable-512.png , badge-96.png , icon.svg).	

Notification-Driven Installation Workflow



3. Core Algorithms, Formulas & Metrics

1. Overall Skill Score Calculation Formula

Calculates the average performance score across all assessed subjects for the logged-in student:

$$\text{Overall Skill Score} = (\sum \text{Assessed Subject Scores}) / (\text{Total Assessed Subjects})$$

2. Subject Assessment Skill Score Formula

Evaluates percentage mastery derived strictly from correct answer accuracy:

$$\text{Skill Score (\%)} = (\text{Correct Answers} / \text{Total Questions}) \times 100$$

3. Standardized Skill Level Thresholds

Score Range (%)	Skill Level Designation	Recommended Action
90% – 100%	Excellent	Advanced topics & peer mentoring eligibility.
75% – 89%	Good	Consolidation exercises & practice quizzes.
50% – 74%	Intermediate	5-Day targeted AI study plan revision.
Below 50%	Needs Improvement	Teacher copilot risk flag & direct remediation.

4. Google Sheets Database Schema (SKILLNEX_DATABASE)

The backend connects to Google Sheets via Google Apps Script Web App API with a dual-mode fallback strategy. If live Google Sheets connection is unavailable, it operates seamlessly in-memory (`USE_DEMO_DATA: true`).

Sheet Name	Header Schema Definitions
Users	<code>user_id, name, email, password, role, city, department, year, xp, level, streak, created_at</code>
Courses	<code>course_id, course_name, subject, description, difficulty, duration, status, created_at</code>
Learning_Progress	<code>progress_id, user_id, course_id, topic, completion, score, time_spent, status, last_activity</code>

Sheet Name	Header Schema Definitions
Assessments	assessment_id, title, subject, topic, question, option_a, option_b, option_c, option_d, correct_answer, difficulty, question_type, created_by, created_at
Assessment_Results	result_id, assessment_id, user_id, score, total_questions, percentage, weak_topic, completed_at
Study_Plans	plan_id, user_id, goal, day, topic, activity, duration, priority, status, created_at
Skills	skill_id, user_id, skill_name, score, level, trend, ai_recommendation, updated_at

5. Complete Codebase Directory Index

SKILLNEX/		
└─ frontend/		
└─┬─ manifest.json	# PWA Web App Manifest file	
└─┬─ sw.js	# Service Worker (Asset Caching & Push Notifications)	
└─┬─ index.html	# Landing Page & PWA Install Controls	
└─┬─ login.html	# 1-Click Role Login (Student, Teacher, Admin)	
└─┬─ register.html	# Registration with Department selection	
└─┬─ student-dashboard.html	# Minimalist Student Hub (Name, Score, Best Subject)	
└─┬─ teacher-dashboard.html	# AI Teacher Copilot & Risk Warning Matrix	
└─┬─ admin-dashboard.html	# Institution & Department Analytics	
└─┬─ study-planner.html	# Single-Subject Difficult Study Planner	
└─┬─ assessments.html	# Subject-Specific Assessment Engine	
└─┬─ skills.html	# Subject-Wise Skill Intelligence Dashboard	
└─┬─ courses.html	# My Courses Catalog	
└─┬─ ollama-ai.html	# Local Ollama AI Workspace	
└─┬─ icons/	# PWA Icon Assets (192, 512, maskable, badge, svg)	
└─┬─ css/ (style.css, landing.css, dashboard.css, components.css, pwa.css, responsive.cs		
└─┬─ js/ (config.js, api.js, auth.js, components.js, pwa-installer.js, dashboard.js, pla		
└─ backend/		
└─┬─ server.js	# Express REST API & PWA Header Server	
└─┬─ GoogleAppsScript.gs	# Google Sheets Web App API Script	
└─┬─ data/knowledgeBase.json	# Knowledge Base Dataset across 12 disciplines	
└─┬─ routes/ & services/	# REST API Routes & Services	
└─ vercel.json	# Vercel deployment & PWA headers configuration	
└─ README.md	# Comprehensive Technical Documentation	

